## Networks and Complexity

# Solution 22-4

*This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB*

**Ex 22.4: Abstract stability analysis [Lvl. 2]**

We place copies of a dynamical system into the nodes of a network and couple them diffusively such that the local within-node Jacobian and the coupling matrix are given by

$$\mathbf{P} = \begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix}, \qquad \mathbf{C} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{1}$$

Compute the master stability function for this system. Describe what the result means for the network.

*Solution*

Using the master stability function trick, we know that we can compute the eigenvalues of the Jacobian as eigenvalues of

$$\mathbf{P} - \kappa\mathbf{C} = \begin{pmatrix} 2 & -3 \\ 3 & -4-\kappa \end{pmatrix} \tag{2}$$

The characteristic polynomial of this matrix is

$$\begin{aligned}
0 &= \lambda^2 + (\kappa+2)\lambda + (1-2\kappa) && (3)\\
0 &= (\lambda + (\kappa+2)/2)^2 + (1-2\kappa) - (\kappa+2)^2/4 && (4)\\
(\lambda + (\kappa+2)/2)^2 &= -(1-2\kappa) + (\kappa+2)^2/4 && (5)\\
(2\lambda + (\kappa+2))^2 &= -4(1-2\kappa) + (\kappa+2)^2 && (6)\\
(2\lambda + (\kappa+2))^2 &= \kappa^2 + 12\kappa && (7)\\
2\lambda + (\kappa+2) &= \pm\sqrt{\kappa^2+12\kappa} && (8)\\
2\lambda &= -(\kappa+2) \pm \sqrt{\kappa^2+12\kappa} && (9)
\end{aligned}$$

and hence

$$\lambda_{1,2} = \frac{-(\kappa+2) \pm \sqrt{\kappa^2+12\kappa}}{2} \tag{10}$$

and

$$M(\kappa) = \frac{-(\kappa+2) + \sqrt{\kappa^2+12\kappa}}{2} \tag{11}$$

The homogeneous state is unstable if there is a $\kappa$ such that $M > 0$. To find the bifurcation points where this happens, we can solve

$$\begin{aligned}
0 &= -(\kappa+2) + \sqrt{\kappa^2+12\kappa} && (12)\\
\kappa + 2 &= \sqrt{\kappa^2+12\kappa} && (13)\\
\kappa^2 + 4\kappa + 4 &= \kappa^2 + 12\kappa && (14)\\
4 &= 8\kappa && (15)
\end{aligned}$$

Hence, the threshold is at

$$\kappa = \frac{1}{2} \tag{16}$$

which implies that the homogeneous steady state is unstable if there is a Laplacian eigenvalue $\kappa \geq 1/2$. In an unweighted network this is always the case if the network contains a link. However, the homogeneous steady state could still be stable on a weighted network if the links are very light.